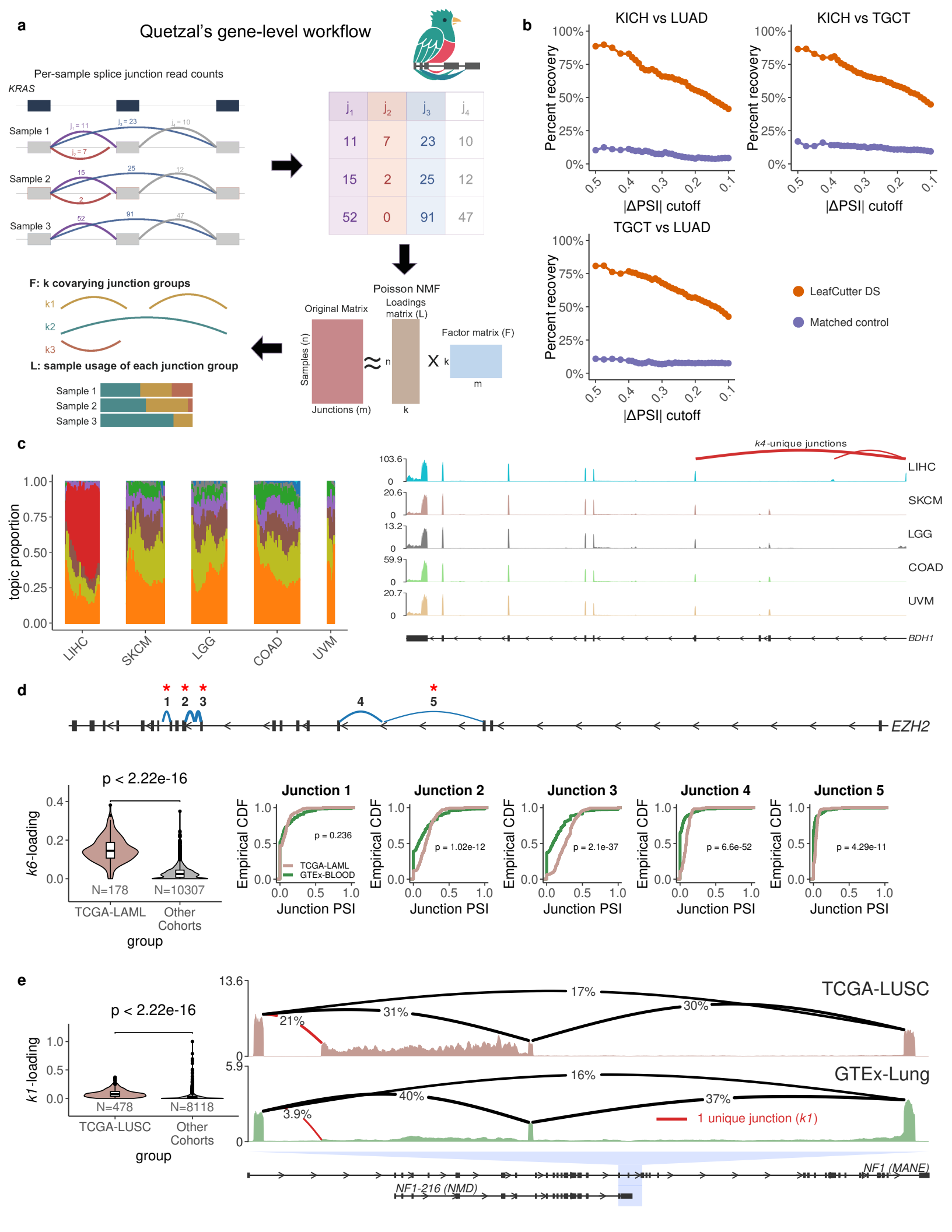

a
Quetzal's gene-level workflow
Per-sample splice junction read counts
KRAS
Sample 1
Sample 2
Sample 3
F: k covarying junction groups
k1
k2
k3
L: sample usage of each junction group
Poisson NMF
Original Matrix
Loadings matrix (L)
Factor matrix (F)
Samples (n)
Junctions (m)
b
KICH vs LUAD
KICH vs TGCT
TGCT vs LUAD
Percent recovery
|ΔPSI| cutoff
LeafCutter DS
Matched control
c
topic proportion
LIHC
SKCM
LGG
COAD
UVM
k4-unique junctions
BDH1
d
EZH2
p < 2.22e-16
k6-loading
N=178
N=10307
TCGA-LAML
Other Cohorts
group
Junction 1
Junction 2
Junction 3
Junction 4
Junction 5
Empirical CDF
Junction PSI
p = 0.236
p = 1.02e-12
p = 2.1e-37
p = 6.6e-52
p = 4.29e-11
TCGA-LAML
GTEx-BLOOD
e
p < 2.22e-16
k1-loading
N=478
N=8118
TCGA-LUSC
TCGA-LUSC
GTEx-Lung
1 unique junction (k1)
NF1 (MANE)
NF1-216 (NMD)